

Nat Anderson

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EXPERIENCE

MICHIGAN TECHNOLOGICAL UNIVERSITY |

UNDERGRADUATE RESEARCH ASSISTANT

Jun 2024 - Apr 2025 | Houghton, MI

- Discussed and tested an interactive Alloy model of the Michigan Clean Slate Act, for use by everyday citizens and/or lawmakers
- Worked on interdisciplinary team of varying experience levels to provide feedback for the model and brainstorm future additions/other models
- Designed and developed an Alloy model of MTU's degree requirements with an interactive framework in mind
- Collaborated to merge this model with a separate academic standing model, created by another team member.

MICHIGAN TECHNOLOGICAL UNIVERSITY |

MATH LEARNING CENTER COACH

Aug 2022 - May 2025 | Houghton, MI

- Assisted other students with homework, exam prep, or general topics related to undergraduate math courses
- Worked in a team of two to manage walk-in hours, and alone for weekly one-on-one appointments
- Ran the learning center walk-in hours alone for the Summer 2024 semester

MICHIGAN TECH RESEARCH INSTITUTE | SUMMER INTERN

May 2023 - Aug 2023 | Ann Arbor, MI

- Used ArcGIS to create training data for a machine-learning model, with the goal of identifying road materials via satellite imagery.
- Developed the backend of a utility program that helped the intern team more efficiently create training data and manage quality assurance.
- Briefly worked for another project, improving a hot loop in an existing program from taking around 1 minute to execute, to taking less than 10 milliseconds to execute on average

PROJECTS

RULES AS CODE | RESEARCH ASSISTANT

Jun 2024 - Apr 2025 | Houghton, MI

- Investigated the usefulness of translating written rules/guidelines into an Alloy model, and creating a interactive interface, or "digital sandbox"
- Most significantly, developed and tested an interactive model for Michigan's Clean Slate Act, allowing users to examine how their own (or fictitious) criminal records would interact with various interpretations of the statute.

MODEL INVENTORY OF ROADWAY ELEMENTS |

SUMMER INTERN

May 2023 - Aug 2023 | Ann Arbor, MI

- Helped train a computer model that could identify where a road is made of asphalt, concrete, or another material via satellite imagery.
- Aimed to create an inventory of all roads in Michigan for MDOT, guided by quarterly meetings with an MDOT team.

EDUCATION

MICHIGAN TECHNOLOGICAL UNIVERSITY

BACHELOR OF SCIENCE IN
COMPUTER SCIENCE AND
MATHEMATICS

Graduated Apr 2025 | Houghton, MI

Cum. GPA: 3.6 / 4.0

SKILLS

PROGRAMMING

5+ years:

Java • Python

3+ years:

SQL • PHP • C/C++ • Alloy

0+ years:

Matlab • JavaScript (React)

TECHNOLOGY

Linux • UNIX • Windows

Git/GitHub • Jira

Microsoft/Google Suites

OTHER

German

3 years • 2 weeks study abroad

Writing • Problem solving • Adaptability

COURSEWORK

GRADUATE

Introduction to Representation Theory

UNDERGRADUATE

Systems Programming

Programming Languages

Model-Driven Software Development

Optimization and Graph Algorithms

Linear Algebra

Combinatorics and Graph Theory

HONORS/AWARDS

Graduated Cum Laude

Norman E. Scholz Award

Deans List 6 semesters